Electrical Power and Energy Systems 134 (2022) 107334

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

Enhancing the transient performance and dynamic stability of microgrid using PI inertia injection controller

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ARTICLE INFO

Keywords:
Derivative Control
RES Penetration
PI Controller
Virtual Inertia
Virtual Damping
Dynamic stability

ABSTRACT

In power sector, the participation of Renewable Energy Sources (RES) is important from an energy, economic, and environmental point of view. But from the grid's perspective, dynamic performance is crucial for deciding the participation level of RES. Currently, the Derivative Control Technique (DCT) based virtual inertia systems are quite popular. However, the use of DCT improves the transient performance but degrades the steady state parameters (settling time and steady-state frequency error). To improve the overall performance, a novel PI-based Inertia Injection controller has been proposed in this paper. This controller advances the Photovoltaic (PV) based virtual inertia model and improves the transient stability without compromising the steady state one. Further, the effectiveness of the proposed controller has been proved by performing the time-domain simulations under different practical scenarios.

1. Introduction

The concept of Virtual Synchronous Generator (VSG) [1] acts as a catalyst in increasing the RES penetration into the power system. The research on VSG is divided into four groups. The First group of researchers (include VSYNCs, ISE Lab, IEPE, KHI Projects [2]), develop and modify the VSG controller. These controllers modulate the output of power inverters and thus provide active power support during contingency [3]. The second group works on the application of VSG in RES dominated power system [4]. The third group mainly focuses on developing different techniques that either work similar to VSG (derivative controlled technique [5]) or optimize the performance of VSG (adaptive inertia [6], alternating inertia [7]). Fourth group discovers various sources (BESS [8], HVDC [9], Ultra-capacitor [10], DC-Link [11], SMES [12]) for extracting the additional energy, required to stabilize the grid during power imbalance periods. The work of this paper belongs to both third and fourth category. In a recent work [13], author has extract the virtual inertia support from solar PV system. The use of DCT for virtual inertia generation, limit the deviation in frequency but increases the time for settling and neutralizing the steady state frequency error. Similar problem of large settling time was solved in [14] by using virtual damping (VD) loop. The VD loop reduces the settling time but it doesn't guarantee zero steady state error [15]. Therefore, virtual damping alone is not a complete solution for these two problems. This research mainly focuses on filling this gap by proposing a complete solution.

Main Contribution: This paper highlights the problem of large settling time and steady state frequency error in virtual inertia systems. Current work majorly contributes in solving these issues by using the PI controller based Inertia Injection Technique. The use of a PI controller at the outlet of DCT based PV and ESS, enhances the frequency performance of virtual inertia systems. This work compares the frequency performance of the PI inertia injection with pre-existed models and promotes the use of PI in DCT based virtual inertia systems.

Setup: The paper is organized in the following pattern; Section I review the previous research work and highlight the research gaps in pre-existed virtual inertia models. Section II performs the frequency response modeling and analyzes the dynamic behavior of the DCT based virtual inertia technique in Section III. Although, the virtual inertia model improves the frequency performance but heavily degrades the steady state response of the microgrid. This problem is solved in section IV; by adding a virtual damping loop and PI controller in the derivative model. Further, the dynamic performance of all the virtual inertia models are compared in Section V and concluded in Section VI.

2. Dynamic modeling

The Microgrid model is a blend of both conventional (e.g. Thermal

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https://doi.org/10.1016/j.ijepes.2021.107334
Received 10 August 2020; Received in revised form 22 January 2021; Accepted 27 June 2021
Available online 6 August 2021
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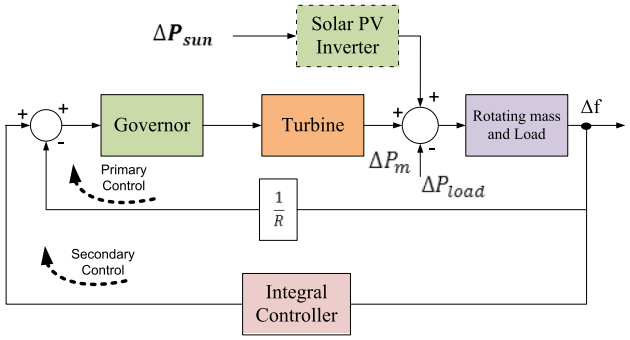


Fig 1. Conventional Microgrid Model.

Power, Nuclear Power) and non-conventional sources of power, used to fulfill the demand of diversified load. This section discusses the dynamic modeling of conventional microgrid and calculates its frequency response.

A. Conventional Microgrid System

The conventional model of microgrid looks like a traditional model, with a difference of the RES system connected as a cumulative active power source. To perform the frequency analysis, a simple low-order linear dynamic model is assembled in Fig. 1.

The conventional dynamic model includes both fast (voltage and rotor angle) and slow dynamics (frequency response). But such models are highly complex and require a large time for deeper analysis. This paper employed a simple model that includes only the slow dynamics of the microgrid system. Neglecting the fast dynamics not only simplify the dynamic model but also reduces the computational time (without interrupting the frequency performance) [15].

B. Mathematical Modeling

Inertia plays a crucial role in deciding the frequency response of microgrid system. The Inertia of the power system is given as;

$$J = \sum_{i=1}^{n} J_i \tag{1}$$

In eq. (1), J is the cumulative inertia of power system that comprises n conventional plants with an individual inertia of J_i. The correlation between kinetic energy and inertia of the system is presented as;

$$K_{kinetic} = \sum_{i=1}^{n} \frac{1}{2} J_i \omega^2 = HS \tag{2}$$

Where, ω represent the angular speed of rotor in rad/sec,H labeled as inertia constant (in p.u. seconds) of the microgrid system with rated power S in MVA. The amount of kinetic energy releases during power imbalance period is governed through swing equation as [16] ;

$$P_m - P_e = \frac{d}{dt}(K_{kinetic}) = \frac{d}{dt}\left(\frac{1}{2} J\omega^2\right) \tag{3}$$

Eq. (3) signifies the fact that the large values of inertia enable the system to make slower change in their kinetic energy. Therefore, during power imbalance events, the higher inertia system shows smaller slope of frequency deviation. From eq. (2) and eq. (3), the rate of change of frequency (RoCoF) can be formulated as;

$$\frac{d\omega}{dt} = \frac{\omega^2(T_m - T_e)}{2HS} = \frac{\omega(P_m - P_e)}{2HS} \tag{4}$$

Here, P_m and P_e is the electrical power generated (equivalent to mechanical power input) and electrical power demanded by the load respectively. Similarly for frequency response, the mathematics involve

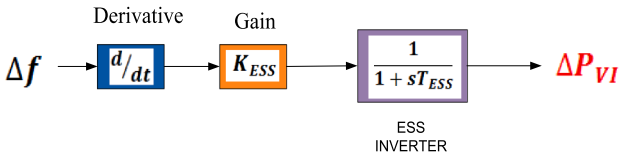


Fig 2. Derivative control technique (DCT) used in ESS model.

is as follows [14];

$$\Delta f = \frac{K_{PS}}{D_L + 2Hs}(\Delta P_m + \Delta P_{solar} - \Delta P_{load}) \tag{5}$$

Where;

$$\Delta P_m = \frac{K_{tr}}{1 + sT_{turbine}}(\Delta P_g) \tag{6}$$

$$\Delta P_g = \frac{K_{gov}}{1 + sT_{gov}}\left(ACE - \frac{1}{R}\Delta f\right) \tag{7}$$

$$ACE(IntegralController) = \frac{K_I}{s}(\beta.\Delta f) \tag{8}$$

$$\Delta P_{solar} = \Delta P_{PV} = \frac{\Delta P_{sun}}{1 + sT_{inv}} \tag{9}$$

Here, Δf represents the frequency deviation that occur due to the inequality between the change in generated power (ΔP_m) and change in load demand (ΔP_{load}) of a microgrid having cumulative inertia constant$H$ with damping coefficientD_L. In eq. (6–8), the change in governor outputΔP_g, droop control ($\frac{1}{R}$) and ACE (Area control error) are acting as primary and secondary frequency controllers respectively. In eq. (9), the change in solar power (ΔP_{solar}) is taken as real power output (as ΔP_{PV}) from solar inverter.

3. Virtual inertia model

In order to sustain the stable operation of microgrid under such an unpredictable environment, virtual inertia is the need of hour. In this section, the virtual inertia models are tested on MATLAB platform.

Setup: The evaluation considers an abrupt load change of 0.2p.u. occurs at 300th of a second. The circuit parameters used in the simulation of virtual inertia methodology is described in (Appendix Table).

A. ESS Based Virtual Inertia Model

In [14], author derived virtual inertia support from battery storage system. The dynamic model of DCT based ESS system is depicted in Fig. 2.

The DCT technique discussed in [5], employs Δf for controlling the active power (ΔP_{VI}) supplied by ESS under load fluctuation events. This extracted power similarly stabilize the microgrid as SG does in traditional power plants [17].

The addition of ESS system upgrades (5) as;

$$\Delta f = \frac{K_{PS}}{D_L + 2Hs}(\Delta P_m + \Delta P_{VI} + \Delta P_{solar} - \Delta P_{load}) \tag{10}$$

Where,

$$\Delta P_{VI} = \frac{K_{ESS}}{1 + sT_{ESS}}\left[\frac{d(\Delta f)}{dt}\right] \tag{11}$$

The ΔP_{VI} represents the virtual inertia support provided by ESS during load fluctuations periods.

B. PV Based Virtual Inertia Model

A novel concept of transforming passive RES into active source of

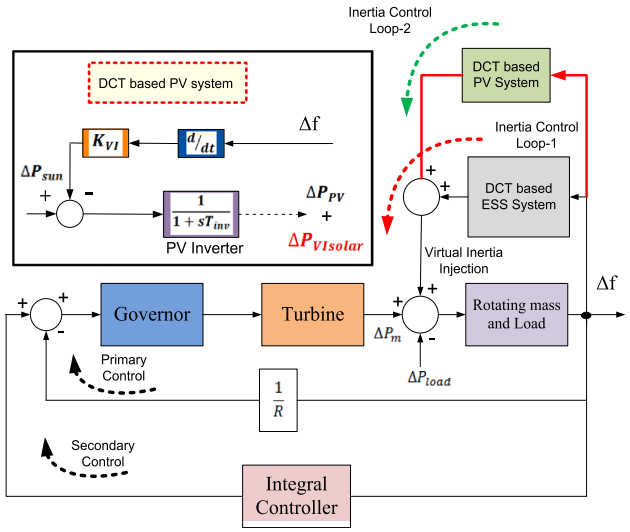


Fig 3. Dynamic Model of PV based Virtual Inertia system [13]

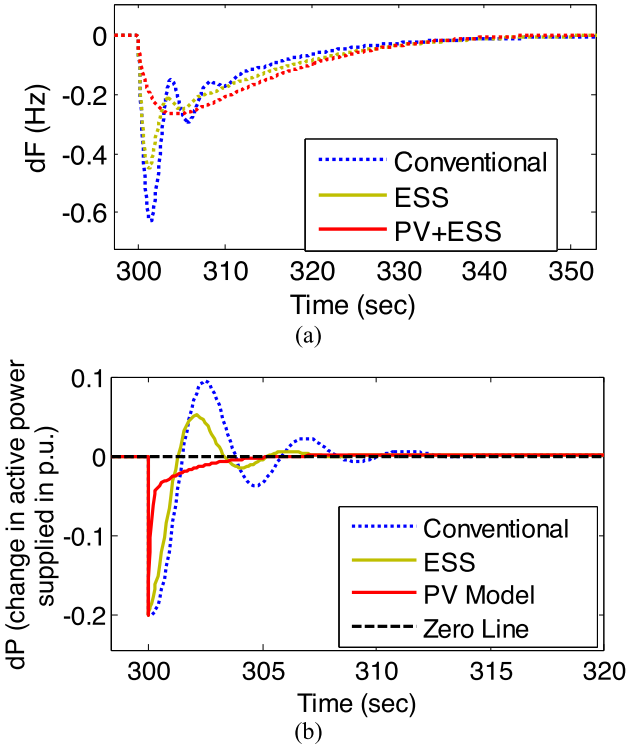


Fig 4. Comparing the effect of virtual inertia technique on (a) frequency deviation (b) change in active power supplied as per load demand.

inertia is proposed in [13]. Under this scheme, the output of PV system is fully controlled by DCT for providing extra inertia support during contingency period. This extra power support is derived from the energy storing elements (DC link capacitors, Batteries, Ultracapacitors) used in PV systems. The complete proposed scheme is shown in Fig. 3. The addition of PV based inertia control (loop-2) modifies eq. (9) as;

$$\Delta P_{solar} = \Delta P_{PV} + \Delta P_{VIsolar} \tag{12}$$

$$\Delta P_{solar} = \frac{1}{1+sT_{inv}}\left[\Delta P_{sun} - K_{VI}\frac{d(\Delta f)}{dt}\right] \tag{13}$$

Where,

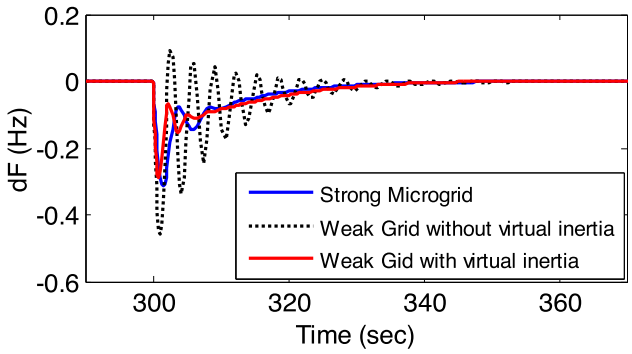


Fig 5. The impact of adding virtual inertia on frequency dynamics of weak microgrid system.

Table 1
Comparing the settling time of different microgrid models.

System	Settling Time	Frequency Deviation
PV Inertia Support	60 Sec	−0.25 Hz
ESS Inertia	40 Sec	−0.44 Hz
Conventional System	20 Sec	−0.63 Hz

$$\Delta P_{PV} = \frac{\Delta P_{sun}}{1+sT_{inv}} \tag{14}$$

$$\Delta P_{VIsolar} = -K_{VI}\frac{d(\Delta f)}{dt}\left[\frac{1}{1+sT_{inv}}\right] \tag{15}$$

Here, ΔP_{solar} is the change in solar power supplied by solar PV systems by DCT controlled solar inverters having T_{inv} as time constant, K_{VI} as control gain of virtual inertia generating system and ΔP_{PV} as inverter's output. The changes occur due to addition of loop-2 (as $\Delta P_{VIsolar}$) .

The dynamics performance reveals that the PV model improves the frequency performance (60% better than conventional and 42% from ESS based microgrid) and stabilizes the system 9 s before the conventional system (see Fig. 4). The positive impact of introducing virtual inertia on weak microgrid system (low inertia system) is simulated in Fig. 5. With virtual inertia support, the weak microgrid is capable of tracing the exact frequency dynamics of a strong microgrid (high inertia system).

The weak microgrid system is simulated by lowering the coefficient of inertia up to 30% (represent 30% RES penetration) from its initial value (H = 21).

C. Issues of large settling time

It is clear from the dynamic performance that the microgrid assisted either with ESS or PV based virtual inertia system, took a large time for settling the frequency fluctuations. This fact is also visible from the Table 1 that the reduction in frequency deviation contrarily increases the settling time of the system.

In case of SG based traditional power system, the load imbalance settles down in a fraction of seconds. The obvious reason is damper winding and Load Damping Coefficient (D_L); provided by frequency dependent loads connected with the system. But in modern power systems, the power converters interfere both on the load and supply side. This interference results in lower D values as the converter application cut out the direct link between supplies and loads. Also, in VSG, there is no such concept of damper winding. This will results in larger settling time and slow frequency dynamics. Some authors [14], introduce the concept of Virtual Damping (VD) in ESS based Virtual Inertia (VI) system. This paper innovatively introduces the concept of VD in PV based

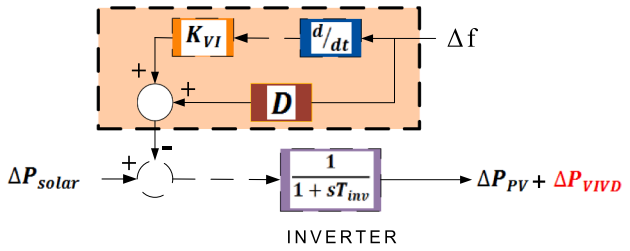


Fig 6. The modified structure of PV based virtual inertia system with virtual damping loop (VD Model).

Table 2
Effect of virtual inertia models on the solar power supplied.

Model	Solar Power (ΔP_{solar})
Conventional	$\frac{\Delta P_{sun}}{1+sT_{inv}}$
ESS	$\frac{\Delta P_{sun}}{1+sT_{inv}}$
PV Model	$\left[\Delta P_{sun} - K_{VI}\frac{d(\Delta f)}{dt}\right]\frac{1}{1+sT_{inv}}$
VD Model	$\left[\Delta P_{sun} - K_{VI}\frac{d(\Delta f)}{dt} - D\Delta f\right]\left[\frac{1}{1+sT_{inv}}\right]$

virtual inertia system and further advanced it by proposing the PI injection controller. The effect of adding VD and PI on system dynamics and transient response will be discussed in the next section.

4. Virtual damping

The role of PV in generating virtual inertia and improving the frequency response has already been discussed in the previous section. This section adds a block of VD and PI in the virtual inertia models and later analyzes their dynamic impact on microgrid performance.

A. Introducing the concept of virtual damping

In PV based virtual inertia model, the RoCoF governs the active power supplied during the contingency period. However, the VD based virtual inertia models use both ROCOF and frequency deviation to control the active power injected (see Fig. 6). This additional frequency controlled loop, works as virtual damping.

The effect of adding the additional loop of VD is formulated in next section.

Mathematically: In previous section (III), (12–15) represents the virtual inertia power supplied by PV system. By introducing the VD concept, (15) is further modified as;

$$\Delta P_{VIsolar} = -\left[K_{VI}\frac{d(\Delta f)}{dt} + D\Delta f\right]\left[\frac{1}{1+sT_{inv}}\right] \tag{16}$$

Here, the D is the Virtual Damping constant. The effect of adding virtual inertia techniques on solar power supplied by PV system is summarized in Table 2.

Result Discussion: The VD model is tested for an abrupt load change of 0.2p.u. (3 MW) with a virtual damping coefficient (D = 1p.u.). The power imbalance curve (see Fig. 7(a)) emphasizes the fact that the VD model quickly responds to change in load demand and neutralizes the power inequality within 2 s (10 sec before conventional and 3 sec before PV model).

In frequency performance, the VD based PV model shows the least deviation (88% better than conventional and 71% better than PV model) after contingency period. This improvement is the result of surplus power supplied by the VD loop. However, it took a large time (240 sec) to completely restore the deviated frequency (equal to the nominal one) during the steady-state. This small deviation is known as the steady-state

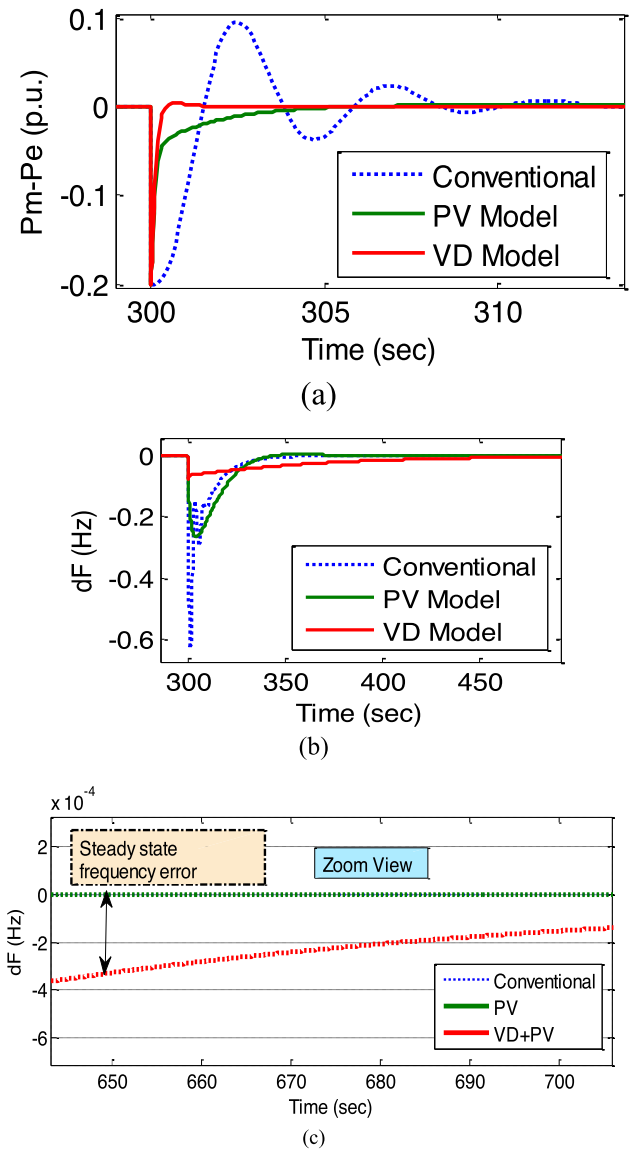


Fig 7. The effect of introducing Virtual Damping on (a) load balancing (b) frequency deviation (c) Zoom view, signify the problem of steady state frequency error.

Table 3
Comparing the dynamic performance of virtual inertia models.

Model	Settling time of load imbalance	Time for zero Frequency error (t_0)
Conventional	12 sec	35 sec
PV Model	5 sec	40 sec
VD Model	2 sec	240 sec

frequency error. The problem of frequency error, represented in Fig. 7 **(b)** and zoomed in Fig. 7**(c)**, compares the frequency performance of the PV model with and without VD. The simulated results are summarize in Table 3;

The time consumed in neutralizing the steady-state frequency deviation is the vital drawback of the VD model. To resolve this issue, a PI controller is applied in the damping loop of PV based virtual inertia model.

B. Introducing PI Inertia Injection Controller

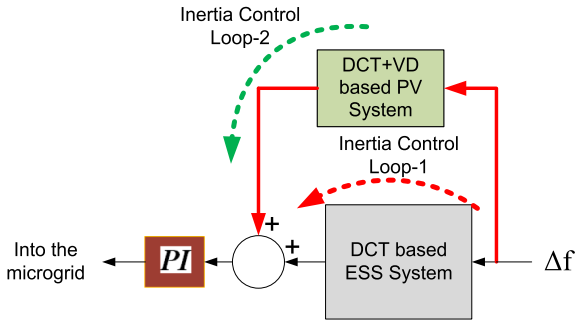


Fig 8. The dynamic model of PI Inertia Injection Controller.

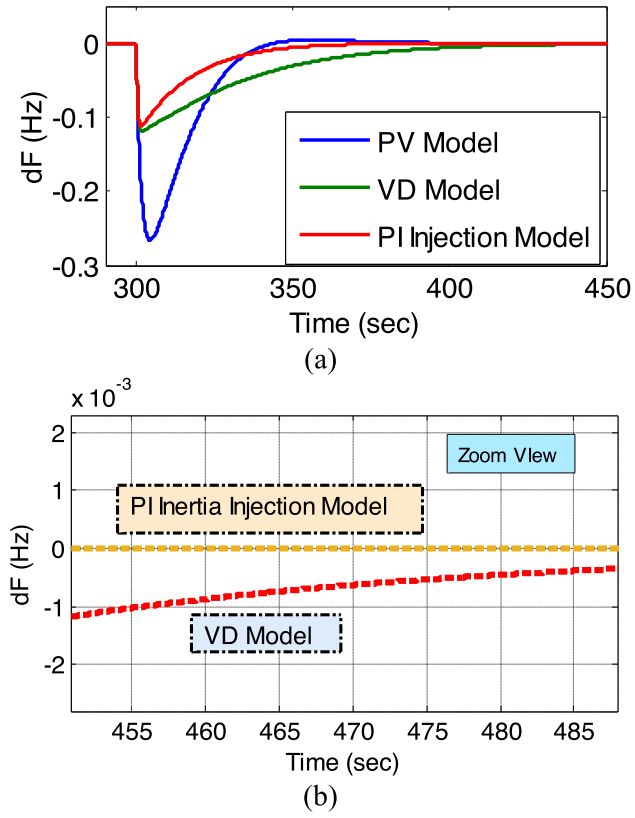


Fig 9. Effect of and PI injection Model on (a) Frequency deviation (b) Zoom view showing the steady state frequency error.

Table 4
Frequency performance of virtual inertia technique.

Models	Maximum Frequency Deviation	Time for zero Frequency error (t_0)
PV Model	0.26 Hz	80 sec
VD Model	0.11 Hz	240 sec
PI Injection Model	0.11 Hz	60 Sec

The PI controller is known for its tendency to quickly settle down the step signal error while operating in steady-state conditions. Therefore, the use of PI in the DCT model reduces the time required to settle and removes the frequency error.

In the proposed model (see Fig. 8), the PI controller is placed at the outlet junction of virtual inertia loops (both ESS and PV). Therefore, the cumulative virtual inertia power injected into the microgrid will pass through the PI controller. The PI operation removes the frequency deviation and enhances both the transient and steady state performance of the microgrid.

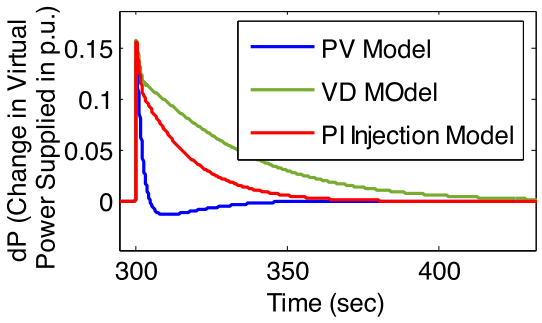


Fig 10. Change in virtual inertia power supplied after abrupt load change at 300th of sec.

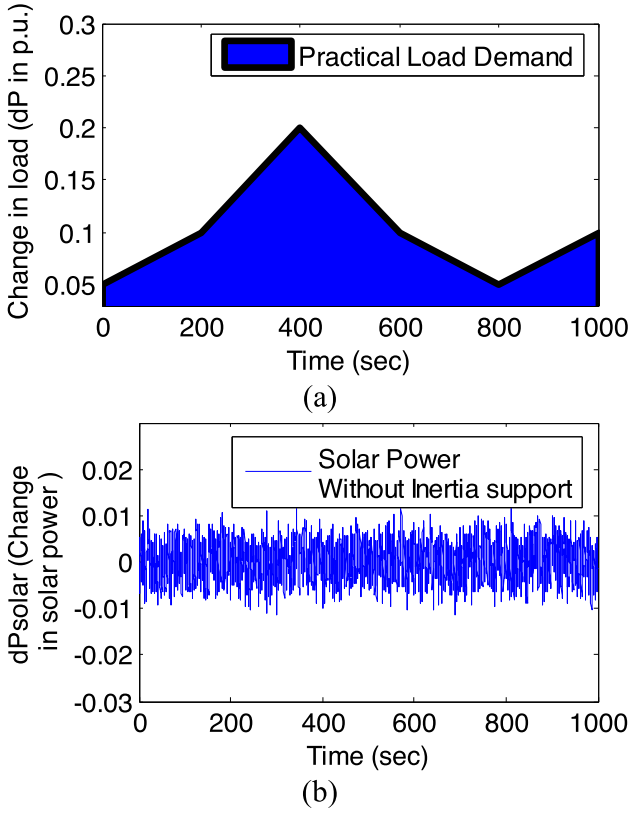


Fig 11. The profile of (a) practical load demand (b) solar power.

Simulation: For simulation, the values of proportional (P) and integral coefficient (I) are selected (via hit and trial method) as 1 and 0.05 respectively. The proposed PI Inertia injection model is tested against abrupt load change of 0.2p.u. and the results are presented in Fig. 9. For analytical comparison, the stats of transient and steady state performance are summarized in Table 4.

In transient performance, the PI model shows the frequency deviation of 0.11 Hz (similar to the VD model), which is 57.7 % better than the PV model (see Fig. 9 (a)). In steady state, the PI model neutralizes the frequency error in 60 sec, which is 20 sec earlier than the PV system and 180 sec than the VD model (see Fig. 9 (b)). The active power injected by virtual inertia models is compared in Fig. 10.

This comparison signifies that the VD model discharges the ESE for a longer duration than the PI model and thus heavily drains out the battery for the same disturbance. This feature adds a positive point to the merits of the PI inertia model. To prove the efficacy of the proposed

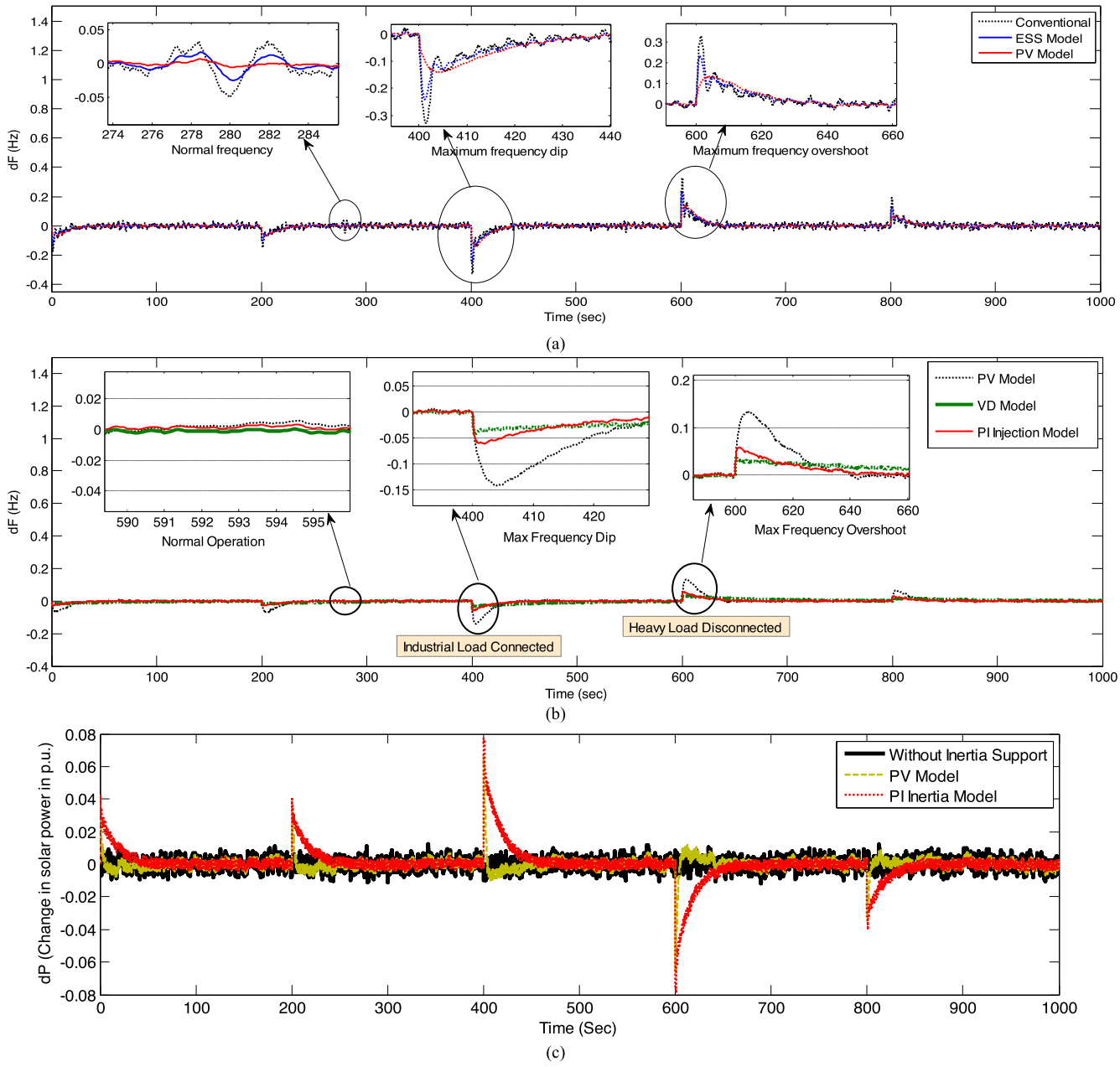


Fig 12. The frequency deviation curve of microgrid under practical load conditions with (a) Virtual Inertia support only and (b) Virtual Inertia and Virtual damping support and (c) change in solar power from MPP.

model, the PI Injection model is simulated against a practical load scenario in the upcoming section.

5. Simulation results

This section tests how effectively the PI model stabilizes the weak microgrid system during fluctuating load demands. For detailed analysis, the section is further divided into different sub-sections.

5.1. Scenario 1: Variable load change

Setup: Here, the microgrid is operated against fluctuating load demand (see Fig. 11 (a)). Due to the replacement of SG via PV system, the net inertia coefficient is taken as 15 (72% of traditional power systems). The output of the solar PV system (taken as RES) operated at its maximum power point (MPP) is shown in Fig. 11 (b).

Table 5
Frequency performance of virtual inertia models under variable load demand.

Models	Maximum Frequency Variation	Time for zero Frequency error (t_0)
Conventional	± 0.64 Hz	35 sec
ESS Model	± 0.46 Hz	35 sec
PV Model	± 0.28 Hz	40 sec
VD Model	± 0.08 Hz	240 sec
PI Injection Model	± 0.12 Hz	40 Sec

Result Discussion: For simplicity, the frequency response of VI based and VD based systems are scoped separately in Fig. 12 (a) and - Fig. 12 (b), respectively. On comparing the frequency response, the PV model dominates in the VI category and PI in VD based systems. The output of the PV system, with and without inertia control, is shown in

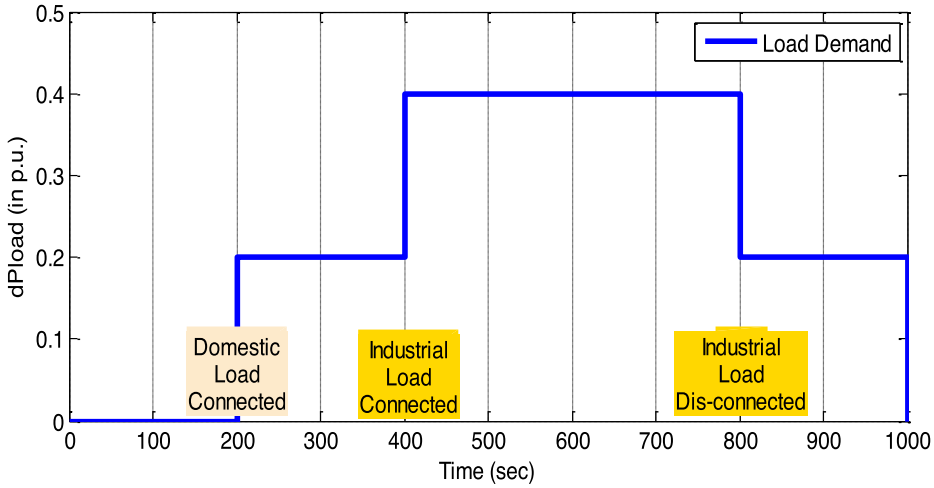


Fig 13. Variable load Profile.

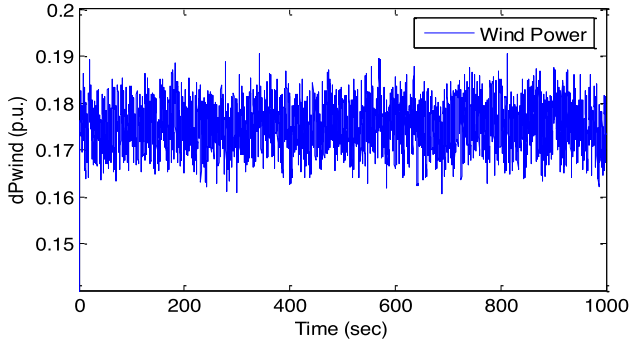


Fig 14. Wind power profile.

Fig. 12 (c). In presence of inertia controllers (PV and PI Inertia Model), the solar power exhibits impulsive variation during the contingency period and thus, actively participates in virtual inertia support (see - Fig. 12 (c)). The results are summarized in Table 5. The performance of virtual inertia models are judged on the basis of time for neutralizing the frequency error and maximum frequency deviation. The conventional system shows minimum t_0 (35 sec) but maximum frequency fluctuation (±0.64 Hz) while VD model, show least frequency deviation (±0.12 Hz) but highest t_0 (240 sec). In mid, the dynamic figures of PI Injection model are quite impressive. The proposed model shows 81% improvement in frequency performance (±0.12 Hz) with a difference of just 5 sec (t_0 of 40 sec) in comparison with conventional one (t_0 of 35 sec).

The simulated results confirm that the VD model excels in transient performance while the PI injection model in neutralizing the steady-state frequency error. However, neglecting the marginal difference in frequency performance, the PI model outcast the VD model while resolving the frequency dynamic issues under variable load conditions.

5.2. Scenario 2: Effect of RES penetration

This scenario examines how the virtual inertia models respond towards variable load demand when a major portion of alternators is replaced by RES.

Load Profile: Here, the practical situation is simulated by continuously changing the load demand (as shown in Fig. 13). To test the peak frequency undershoots a domestic load of 0.2p.u. is applied at 200th of sec and an industrial load (0.2p.u.) at 400th of sec. Similarly, for the overshoot performance, the industrial load is disconnected along with the connection of the wind power plant (WPP) at 800th of sec. The output of the wind power plant is shown in Fig. 14.

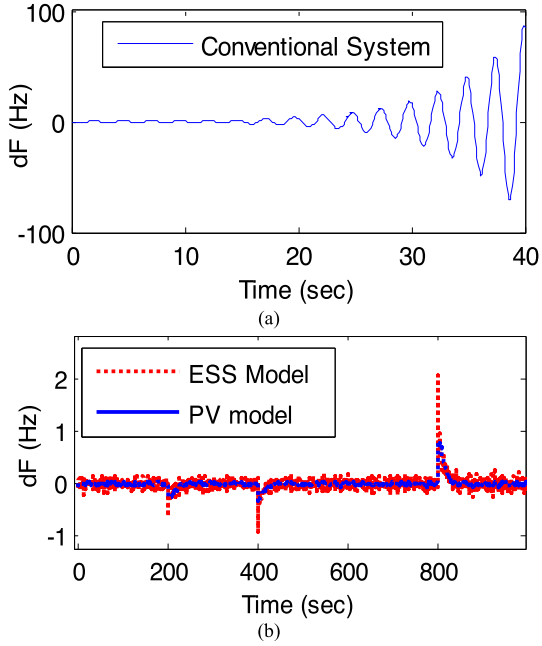


Fig 15. Frequency performance of (a) Conventional and (b) Virtual inertia based microgrid.

Penetration effect: The penetration of PV into the microgrid directly affects the dynamic performance and weakens the system, inertially. To simulate a weak microgrid, 80 % of SGs are replaced via non-inertial PV systems. As a result, the value of H was reduced to 20 % (H = 4.2 sec) of its traditional value (H = 21 sec). Similarly, the use of converters while interfacing the load affects the damping coefficient of the system. To incorporate this effect, the damping coefficient of the system is reduced to 20 % ($D_L = 0.28$) of its conventional value ($D_L = 1.2$).

Results Discussion: The simulation results imply that in absence of virtual inertia support, the conventional microgrid fails and may lead to severe consequences like blackout (see Fig. 15 (a)). Therefore, the conventional microgrid doesn't support the deep RES penetration. Considering the case of virtual inertia models, both ESS and PV systems supports the grid, virtually. The ESS model shows a frequency fluctuation of ± 0.2 Hz (during 0 to 200 sec) and −0.9 Hz at 400th of sec (industrial load connection). In comparison, the PV model shows a frequency variation of ± 0.03 Hz (normally) and −0.3 Hz at 400th of sec

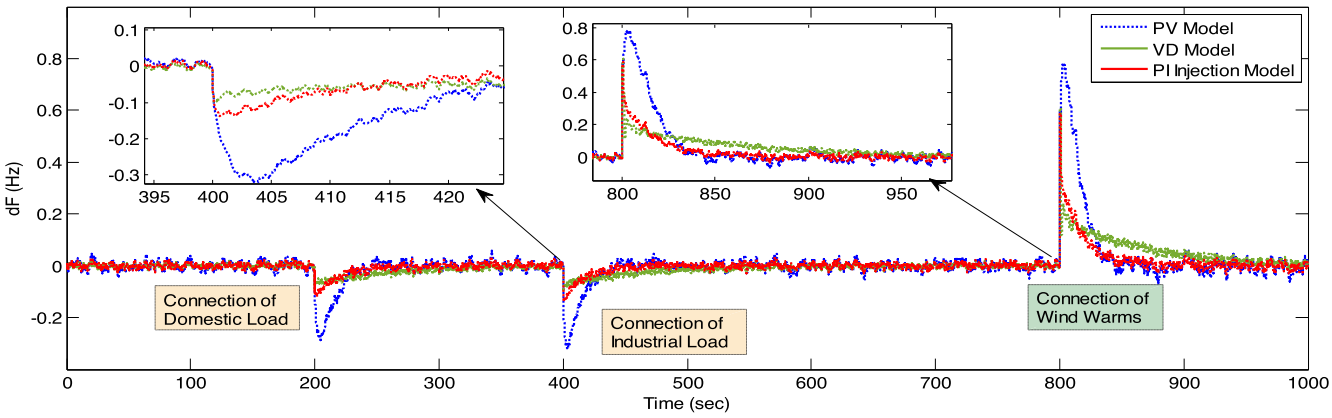


Fig 16. Frequency performance of virtual inertia & virtual damping based microgrid system.

Table 6
Summary of transient & steady state performance.

Models	Maximum Frequency Undershoot	Maximum Frequency Overshoot	Time for zero Frequency error (t_0)
ESS Model	0.91	2	35 sec
PV Model	0.31	0.78	40 sec
VD Model	0.1	0.61	240 sec
PI Injection	0.14	0.59	40 Sec

(see Fig. 15 (b)).

They both supply the required amount of inertia in virtual form and restrict the undershoots, within the permissible range (±1.5 [18]). However, during wind farm connection at 800th of sec, the ESS model fails in limiting its peak frequency overshoot (2 Hz) while the PV model stabilizes the microgrid with an overshoot of 0.8 Hz.

Although the PV based virtual inertia system passes the test of stability, but its time for settling and neutralizing the steady-state frequency error needs further improvement. As a solution, the virtual damping models (VD and PI injection) are used to enhance the dynamic performance of the microgrids (see Fig. 16). The simulation results conclude that during load connection; the VD model limits the frequency variation up to −0.1 Hz and + 0.6 Hz during the wind farm connection. But, it takes a large time to resolve the frequency errors. In comparison, the PI injection model shows the frequency variation, similar to the VD model. But, neutralizes the frequency error earlier than VD based microgrid (see table 5).

The frequency performances of other virtual inertia techniques are summarized in Table 6. It is clear from the comparison table that the difference of t_0 in PI injection and VD model is of 200 sec. It is a large time gap, as per the dynamics of the microgrid concerned. Therefore in comparison with the VD model, the PI injection model is superior in transient performance while equivalent in limiting the frequency deviation. Hence, it is better to select the PI Injection model (over VD or PV model) for high-quality transient and steady state performance during the contingency period.

6. Conclusion

The depleted fossil fuels, increasing load demand, and continuous downfall in the environmental quality predict that the RES plays an important role in current and future microgrid systems. In this paper, the work of RES based virtual inertia technique has been further enhanced by introducing a PI Inertia Injection controller. For validation, the PI-based microgrids are tested under the scenarios, ranging from practical load conditions to deep RES penetration. The use of PI advances the conventional by 81%, ESS by 67.3 %, and PV by 57.1% in their transient performance. These results confirm that the PI-controlled PV system is quite effective in improving both the transient and steady state performance of the microgrid system. Thus, the usage of PI for injecting virtual inertia encourages the RES participation in future microgrid system. Further, an extensive research is required to optimize the parameters of the PI controller and explore its advanced application in the future power system.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

	Microgrid Parameters	Values
Primary Control Unit	Time constant of governor, T_{gov}(s)	0.4
	Gain of speed governor, K_{gov} (s)	10
	Time constant of turbine, $T_{turbine}$ (s)	0.5
	Gain of turbine, K_{tr} (s)	0.05
	Droop control, R (Hz/ p.u. MW)	1
	Bias factor, (β) [p.u. MW/Hz]	1
	Integral Control variable gain, K_I(s)	0.09
Virtual inertia control (ESS)	Control gain of virtual inertia of ESS, K_{ESS} (s)	1.6

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(*continued*)

	Microgrid Parameters	Values
	Time constant of Inverter based ESS, T_{ESS} (s)	10
	Maximum capacity of ESS, $P_{ESS\ max}$ (p.u. MW)	0.4
	Minimum capacity of ESS, $P_{ESS\ min}$ (p.u. MW)	−0.4
Virtual inertia control (RES)	Control gain of virtual inertia of PV system, K_{VI} (s)	1
	Time constant of inverter-based PV system, T_{inv} (s)	0.5
	Virtual Damping (VD) coefficient	1
	Integral Coefficient of PI Controller (I)	0.5
	Proportional Coefficient of PI controller(P)	1
	Maximum inertia support by RES, $P_{RES\ max}$ (p.u. MW)	0.4
	Minimum inertia support by RES, $PRE_{S\ min}$ (p.u. MW)	−0.4
Microgrid control unit	Power System gain constant, K_{ps} (s)	100
	Inertia constant of the system, H (p.u. MWs)	15
	Inertia constant of the traditional system, H (p.u. MWs)	21
	Damping constant of the system, D_L (p.u. MW/Hz)	1
	System Base (MW)	15

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijepes.2021.107334.

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